Uncoupled performance: Exploring trait-specific responses to predict species' sensitivity to temperature change

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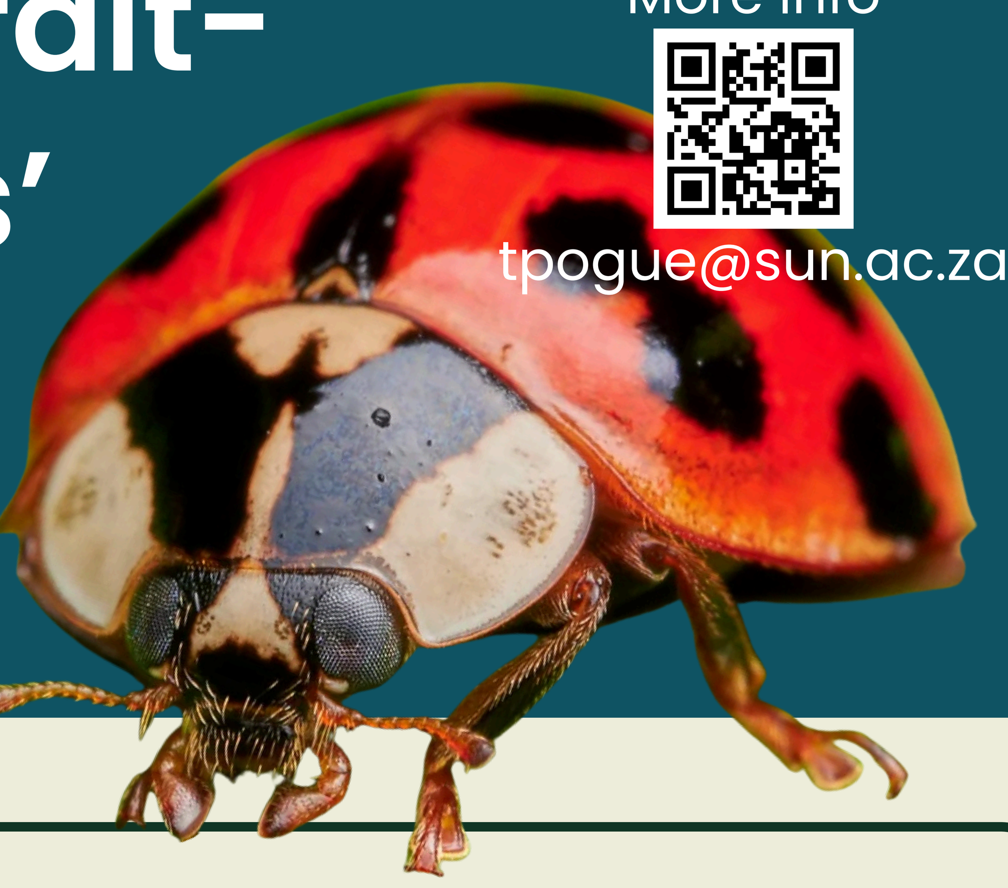


BACKGROUND

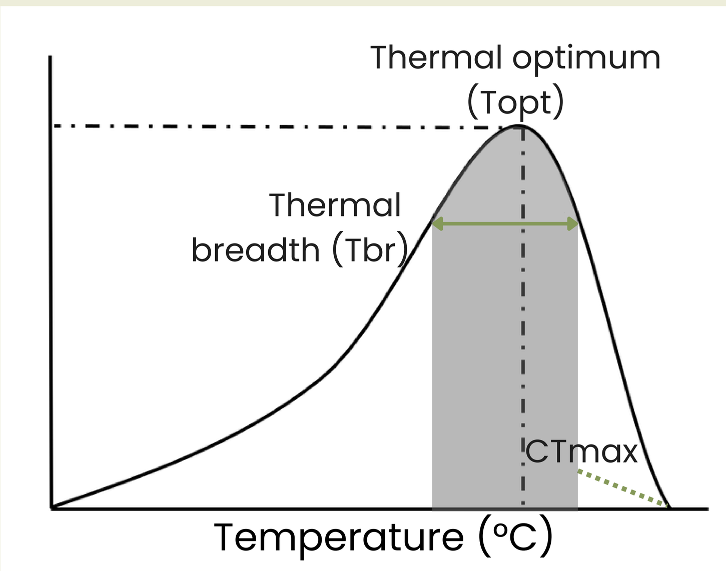
- Thermal performance curves (TPCs) are used to predict species thermal sensitivity, often relying on one trait as a proxy for fitness
- However, traits vary in their thermal sensitivity
- Variation among trait TPCs is rarely quantified within the same population
- This may lead to inaccurate estimates of climate vulnerability and realised performance

METHODS

F_0 → F_1 → F_2 *Harmonia axyridis* → Broods split across temperatures → TPCs tested (Lab): 5–48°C

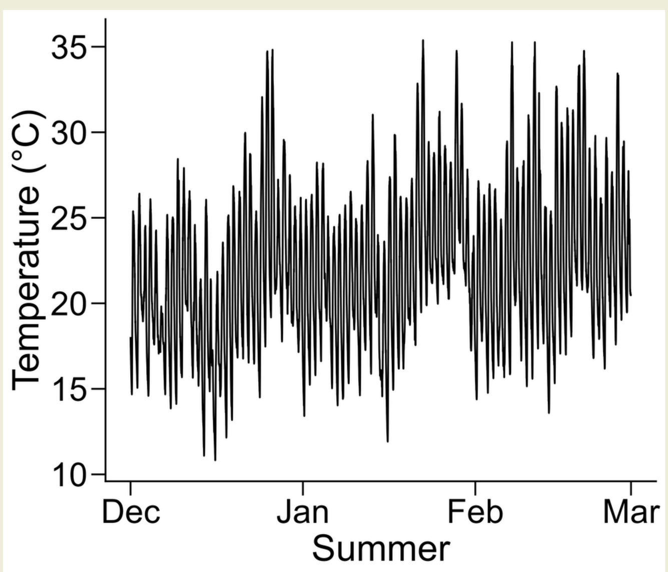
Lab

TPC data

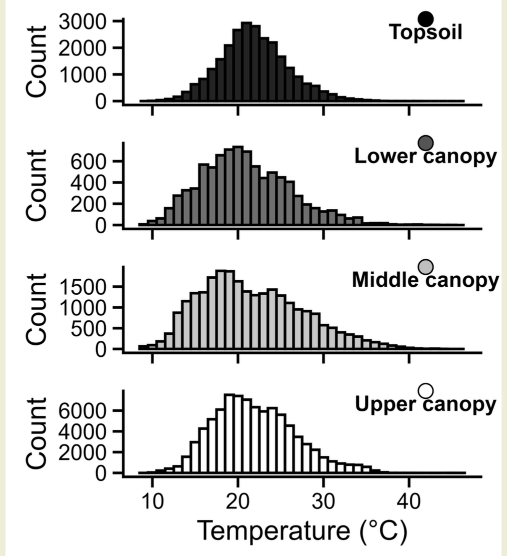


Field

Temperature timeseries

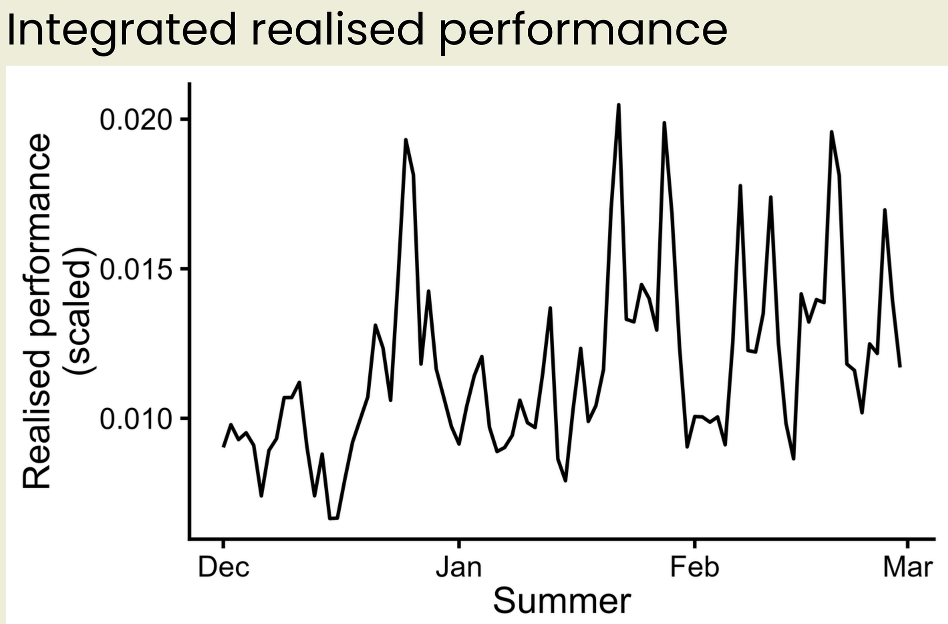


Microsite temperature distribution



Lab x Field

Integrated realised performance



$$\langle w \rangle = \int_{T_{min}}^{T_{max}} w(T)PDF(D,T)dT$$

- $\langle w \rangle$ = performance
- $w(T)$ = performance at temperature T
- $PDF(D,T)$ = probability density of temperature distribution D at T

FOR MORE INFORMATION ON THE METHODS SCAN THE QR CODE

How do TPC differences between traits effect realised performance estimates?

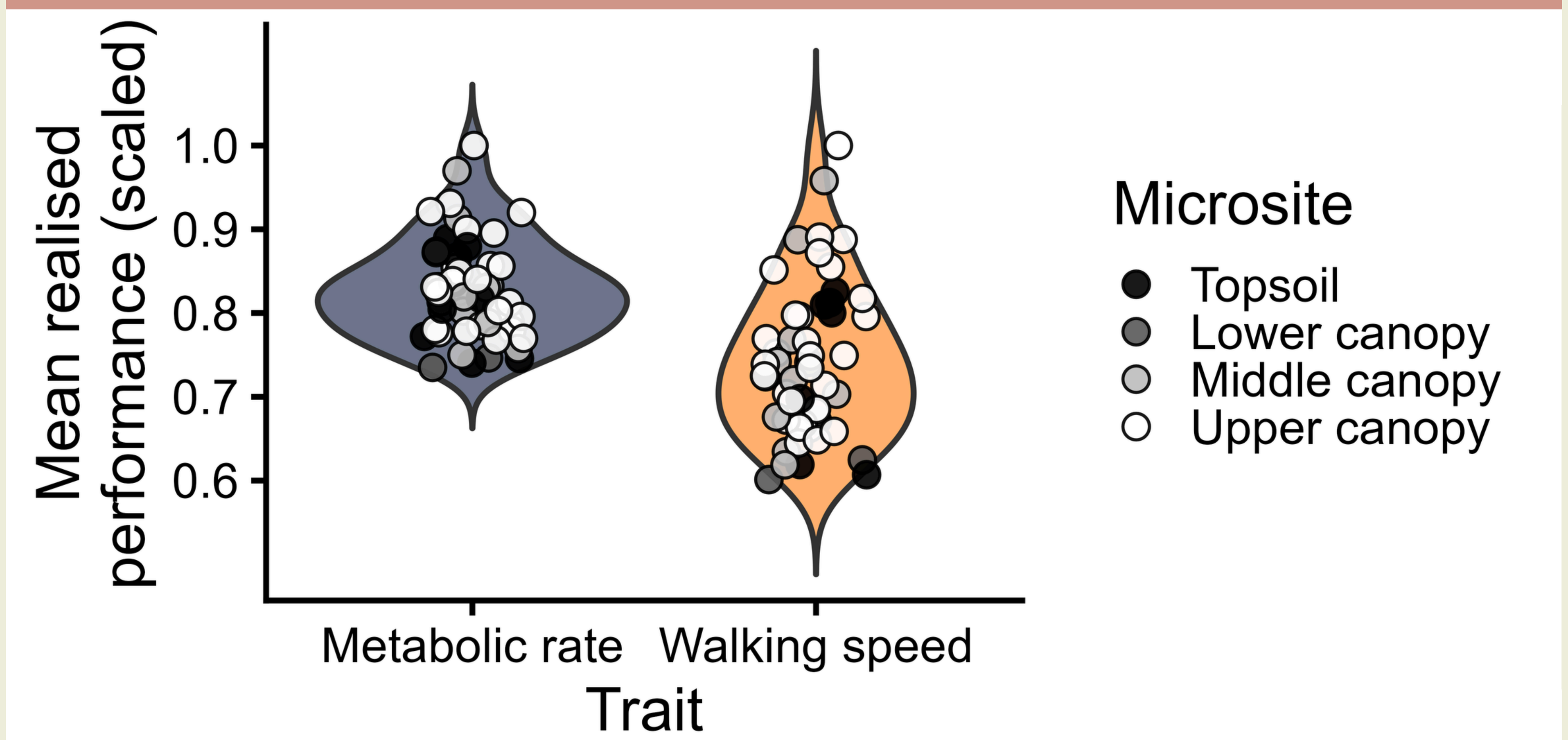


Fig 2. Mean scaled realised performance of metabolic rate and walking speed across microclimates used by *H. axyridis* in Stellenbosch, South Africa during summer.

KEY RESULTS

- **TPCs differ** amongst traits (Fig. 1)
- Natural **temperature variability** leads to **trait-dependent realised performance** (Fig. 2)
- Species-relevant **microclimate scales impact performance** estimates (Fig. 2)
- Vulnerability assessments should incorporate trait differences and fine scale thermal measurements

How do TPCs differ amongst traits in the same population?

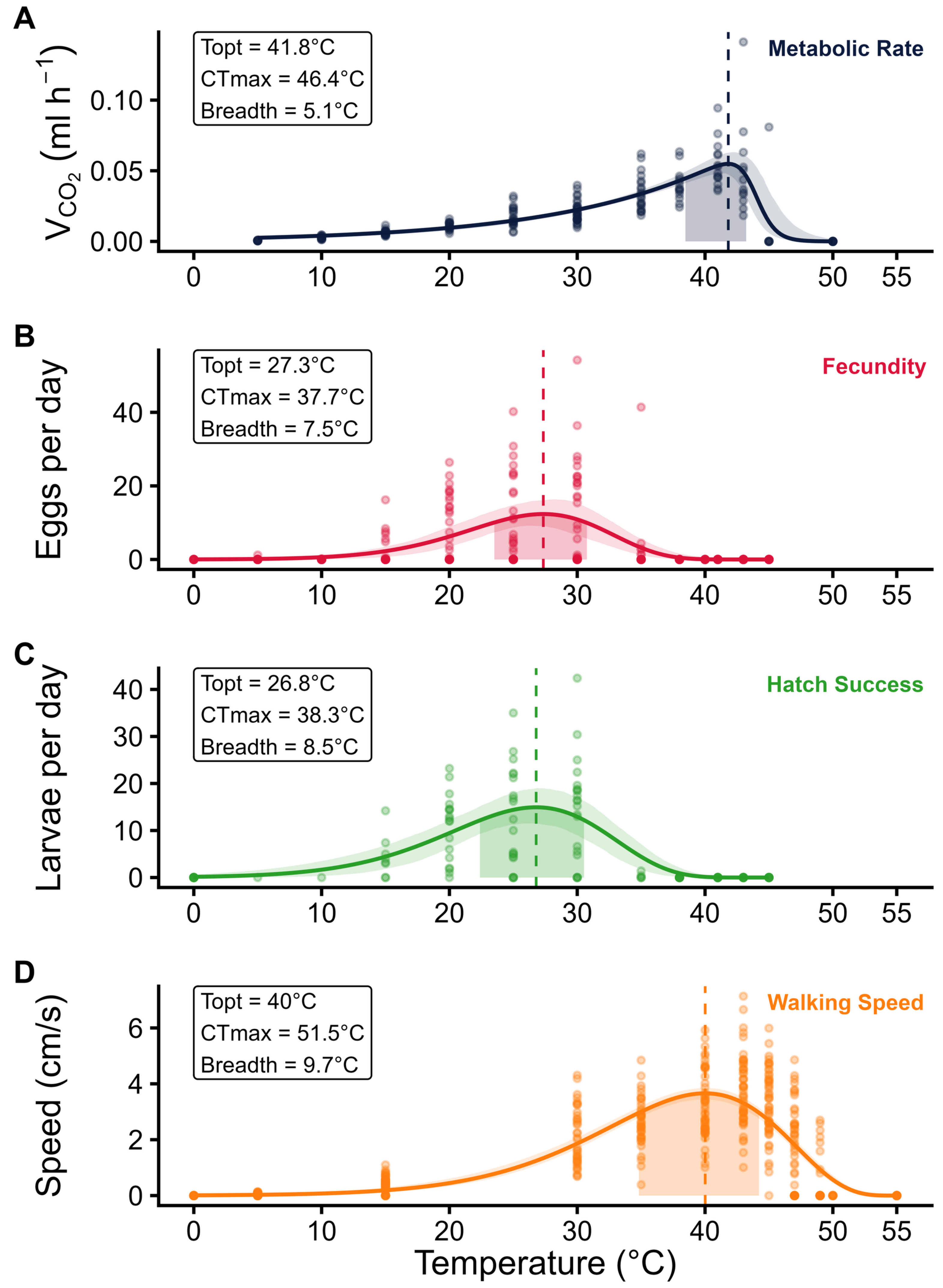


Fig 1. Thermal performance curves for *H. axyridis*. Shaded regions indicate thermal breadth (≥80% of maximum performance). Curves represent the best-supported model fit based on lowest AIC scores from data collected across 5–48°C (n = 15 per temperature), with key thermal parameters indicated for each trait.

NEXT STEPS

- Link enzyme activity & performance TPCs
- Refine realised performance estimates using 2-year timeseries of thermal data
- Incorporate exposure (behaviour & relevant temperature variation) into realised performance
- Determine the best TPC-based predictors for fitness

British Ecological Society · Entomological Society of Southern Africa · Stellenbosch University · C·I·B Centre for Invasion Biology · NRF National Research Foundation · CapeNature · Ernst & Ethel Eriksen Trust

This work was supported by the National Research Foundation, the Centre for Invasion Biology, the Ernst and Ethel Eriksen Trust, and Stellenbosch University. Travel funding was provided by the British Ecological Society and the Entomological Society of Southern Africa. We thank Bridget O'Connor for assistance with the walking speed thermal sensitivity experiments.